

Data Mining and Machine Learning

Multimodal Emotion Recognition

A classifier based on EEG and Eye Movement recordings

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Project outline

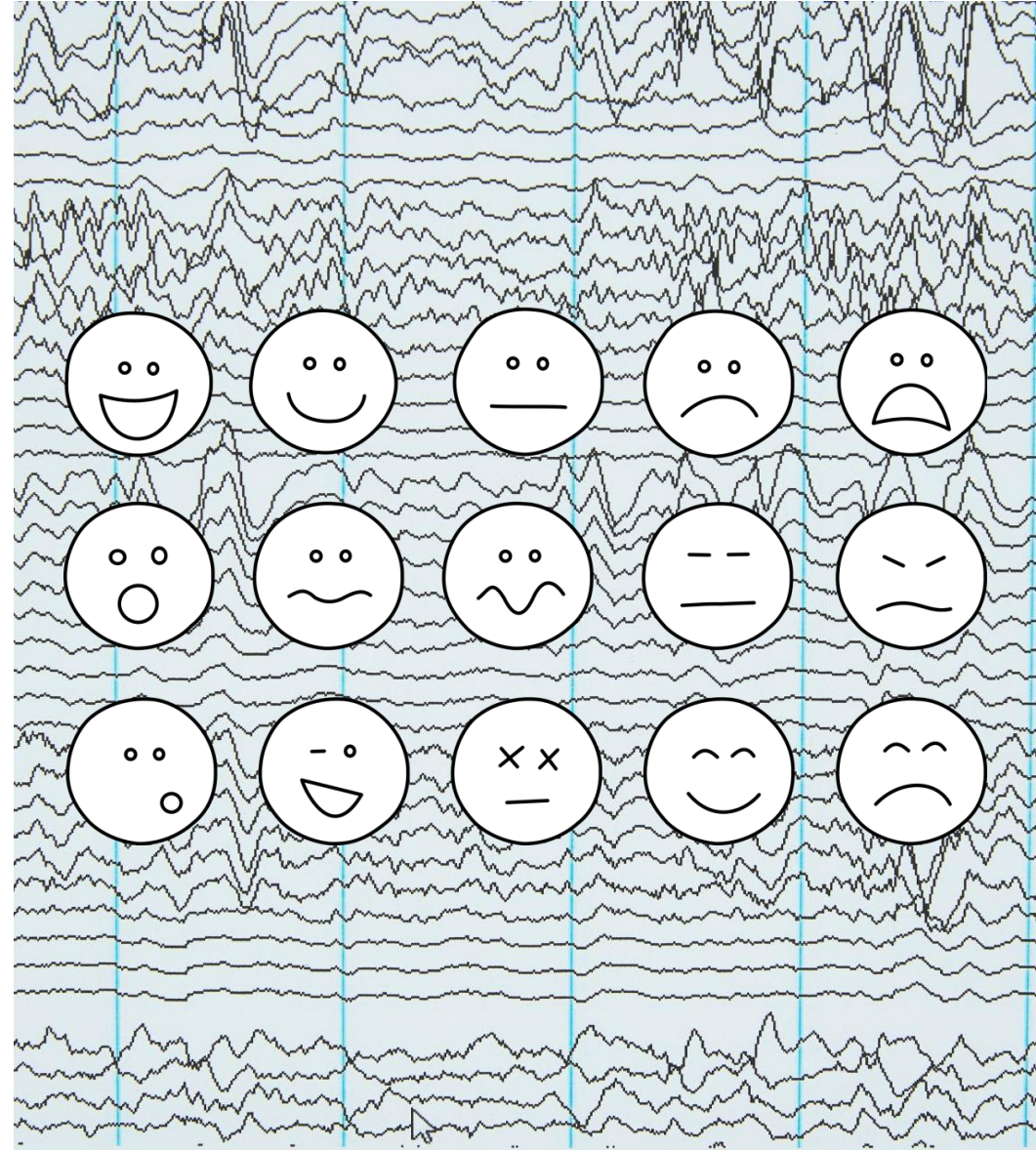
This project aims to develop a **classifier** for **recognizing emotional state** of an individual, discriminating between neutral state, happiness, sadness and fear. To this purpose, a promising approach has been shown to be the combination of EEG recordings and external behaviors -in this case eye movements features- to enhance classification performance.

1. Emotion Recognition
2. The SEED IV dataset
3. EEG only classification
4. Eye movements only classification
5. Modality fusion: Early Fusion
6. Modality fusion: Late Fusion
7. Models Comparison
8. Conclusions



Emotion Recognition

- Emotions play an important role in our daily lives, specifically in human-human interactions: the emotional state of an individual modulates the way information is shared, affecting several external behaviours like posture, tone of voice, facial expressions and gestures, which in turn can alter the meaning of a sentence.
- Moreover, the ability to recognise one's affective state has been shown to play an important role in learning, and is used to evaluate children development.
- Introducing affective awareness in human computer interactions is currently the object of interest of many fields that go beyond psychiatry, with applications that encompass robotics, neuroscience, marketing, education, entertainment.



SEED- IV Dataset

The **SEED-IV** dataset is an evolution of the original SJTU Emotion EEG Dataset (SEED), provided by the BCMI laboratory (Brain-like Computing and Machine Intelligence) in Shanghai.

It provides **EEG signals** collected with the **62-channel** ESI NeuroScan System and **eye movement features** recorded by SMI eye-tracking glasses, which makes it a well-formed multimodal dataset for **emotion recognition**.



- The experiment is organized in **3 sessions**, performed in different days, each consisting of **24 trials** per subject. This project will exploit only two sessions.
- During a trial, each of the 15 subjects (healthy, right-handed aged between 20 and 24 years) is shown a different clip which induces one of the 4 target emotions: happiness, sadness, fear, neutral state.
- At the end of each experiment, the subjects were asked to assess their emotions when watching the film clip with keywords of emotions: neutral, sad, fear, and happy , labeled respectively 0, 1, 2, and 3 .

Link to the dataset: <https://www.kaggle.com/datasets/phhasian0710/seed-iv>

EEG Dataset: Preprocessing and Feature Extraction

Preprocessing and feature extraction was carried out following Zheng et al. proposed methodology : each recording was epoched with 4 seconds windows, bandpass filtered between 1-45 Hz and afterwards PSD and Differential entropy (DE) were extracted for each of the 5 bands of interest: delta(1~4 Hz), theta: (4~8Hz), alpha(8~14 Hz), beta (14~31 Hz) and gamma(31~45 Hz)

EEG DATASET ORGANIZATION

The **raw EEG signals** of each subject for each trial are provided in a 62xN matrix, where the number of rows refers to the channels and N is a variable number of samples which depends on the duration of the trial considered.

Each recording is sliced into 4-second nonoverlapping segments. Each segment is regarded as one data sample.

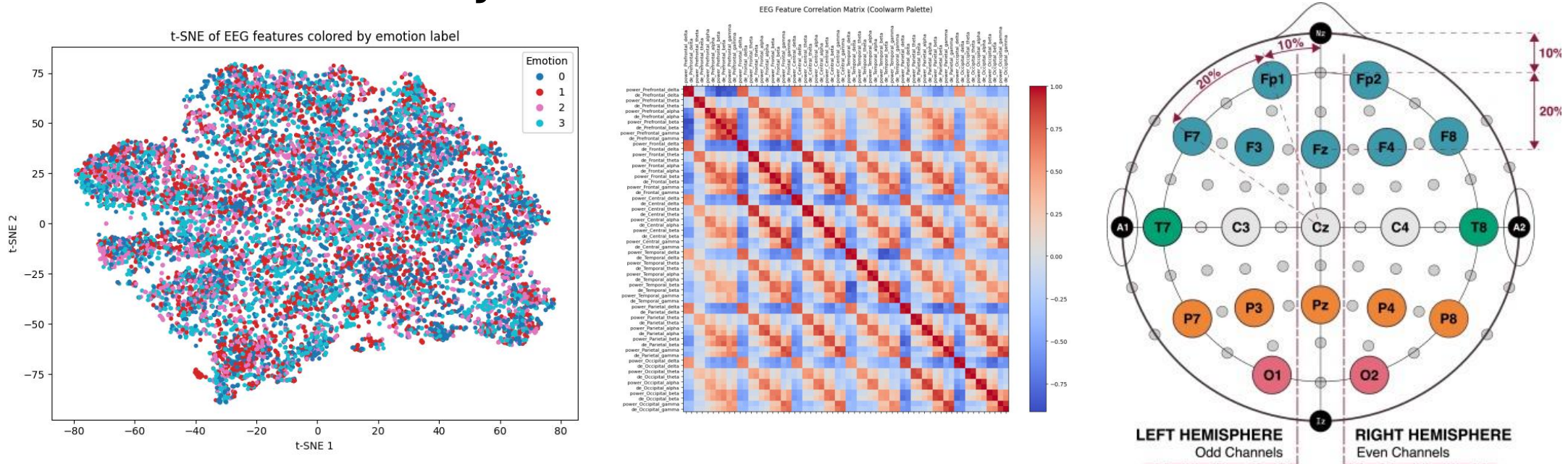
$$\text{PSD} = E[x^2]$$

$$\text{DE} = - \int_{-\infty}^{\infty} P(x) \ln(P(x)) dx$$

	subject	trial	label	power_Prefrontal_delta	de_Prefrontal_delta	power_Prefrontal_theta	de_Prefrontal_theta	power_Prefrontal_alpha
0	1	1	1	5.419316	2.788496	0.965721	1.348985	0.285273
1	1	1	1	5.334861	2.772089	1.120173	1.496136	0.199796
2	1	1	1	6.035457	2.896235	0.885443	1.264420	0.135681
3	1	1	1	6.615760	2.988065	0.577991	0.838034	0.115213
4	1	1	1	6.174299	2.918550	0.369509	0.387299	0.153445
...	...	...	...	...	...	...	...	...
12760	15	24	3	1.937747	1.758818	0.719483	1.055710	0.447947
12761	15	24	3	5.407810	2.786447	1.336944	1.675631	0.183453
12762	15	24	3	4.383929	2.569521	0.783946	1.137337	0.606540
12763	15	24	3	5.543900	2.810762	1.359158	1.691466	0.136645
12764	15	24	3	4.787874	2.660478	1.019307	1.403549	0.338062

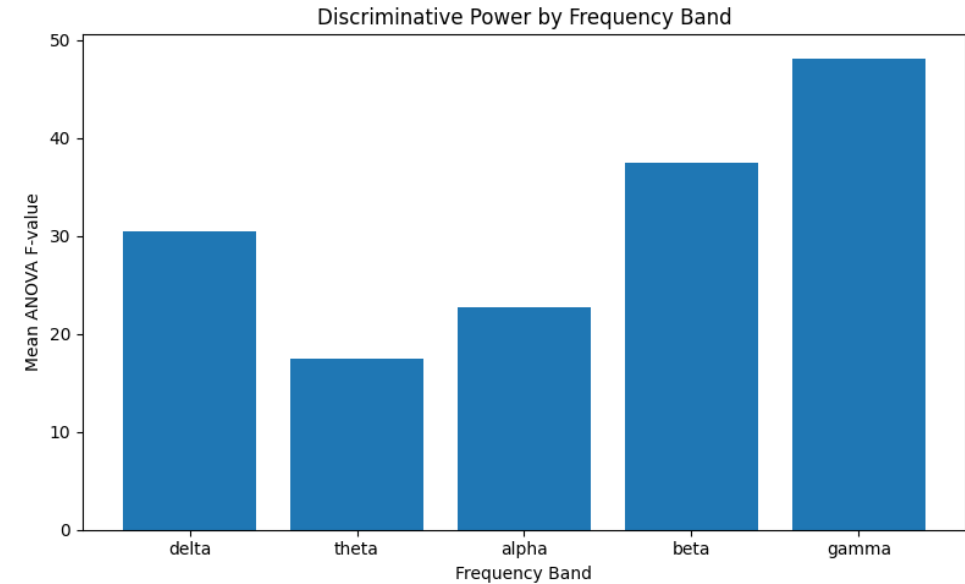
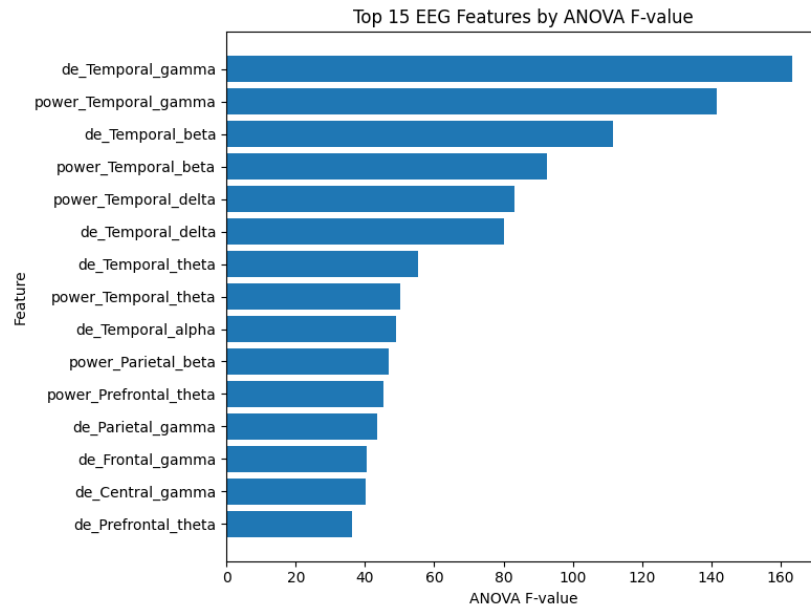
The final features are in the format **feature_i_channel_j_band_k**

EEG Dataset: analysis



- As a preliminary step, EEG channels were grouped into six regions of interest—Prefrontal, Frontal, Central, Parietal, Temporal, and Occipital—to reduce feature redundancy
- No clear clusters in t-SNE embedding → high feature dimensionality and overlap obscure separable patterns
- Correlation matrix shows strong diagonal and near-diagonal correlations → many features highly interrelated
- Visual inspection alone cannot pinpoint the most discriminative channels or bands

EEG Dataset: analysis



- Zheng et al. indicate the beta and gamma bands as being the most informative
- The computed ANOVA F-scores confirm this claim, leading us to ultimately selecting the 3 bands with the highest scores: beta, gamma and delta.

EEG Classification



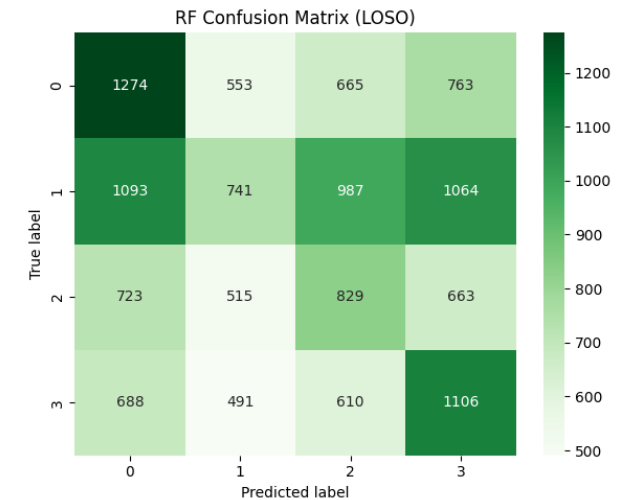
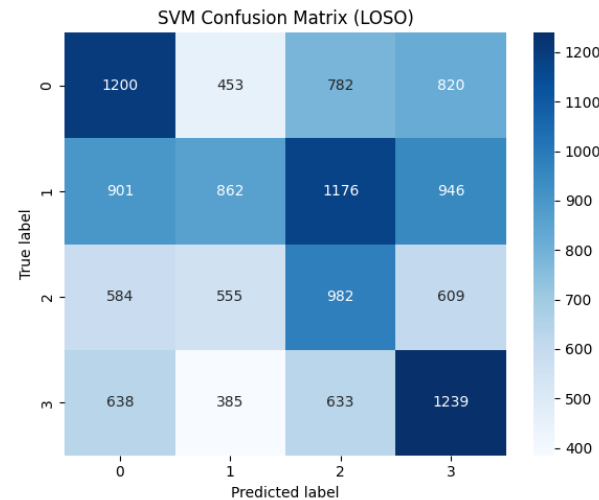
PHASE 1: HYPERPARAMETER TUNING & TRAINING

- LOSO Cross Validation ensures subject independent validation
- Definition of search space
- For each set of values proposed, a LOSO CV is performed to evaluate the parameters
- After the best parameters have been found, the model is retrained on all session one data
- SVM and RF were evaluated

Accuracy: 0.3355

	precision	recall	f1-score	support
0	0.3611	0.3687	0.3649	3255.0000
1	0.3823	0.2219	0.2808	3885.0000
2	0.2748	0.3597	0.3116	2730.0000
3	0.3428	0.4280	0.3807	2895.0000
macro avg	0.3403	0.3446	0.3345	12765.0000
weighted avg	0.3450	0.3355	0.3315	12765.0000

RF LOSO accuracy: 0.309



EEG Classification



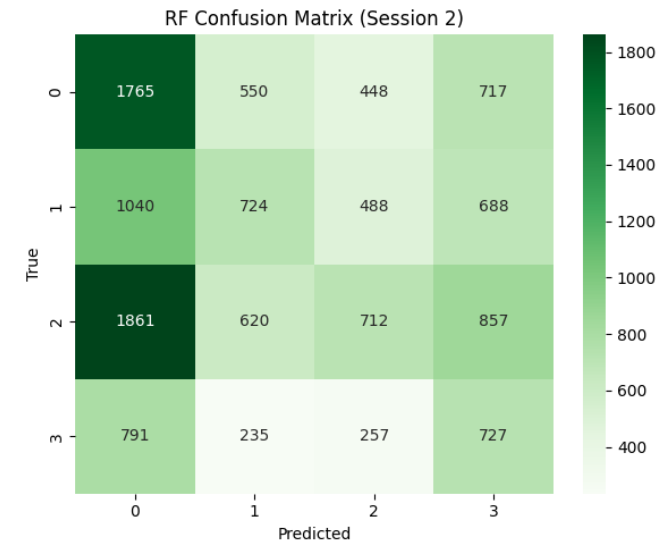
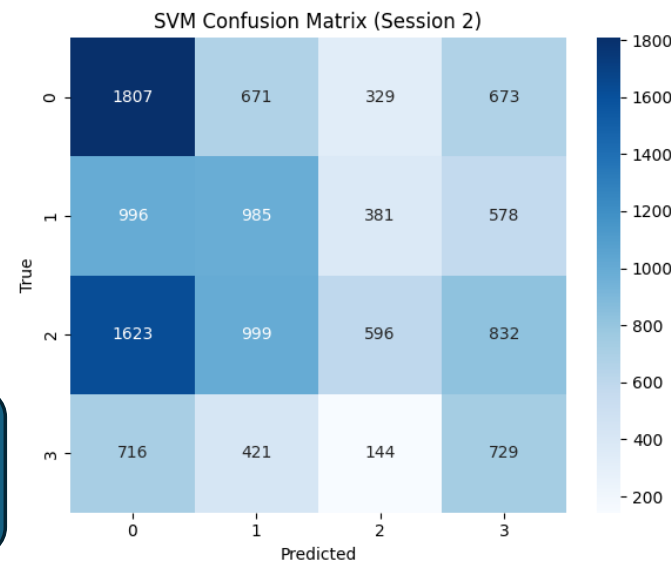
PHASE 2: INDEPENDENT TEST ON SESSION 2

- Data from same subjects, different recording session
- Features preprocessed & selected identically to training
- This ensures no leakage and is a good way of verifying robustness of the model
- Overall accuracy $\approx 33\%$ (vs. 25 % chance-level for 4 classes)

SVM test-on-session2 accuracy: 0.330

	precision	recall	f1-score	support
0	0.35	0.52	0.42	3480
1	0.32	0.34	0.33	2940
2	0.41	0.15	0.22	4050
3	0.26	0.36	0.30	2010
accuracy			0.33	12480
macro avg	0.34	0.34	0.32	12480
weighted avg	0.35	0.33	0.31	12480

Class0 (neutral state): most reliably detected
Class 2 (fear): rarely recognized



EEG Classification



PHASE 2: INDEPENDENT TEST ON SESSION 2

Eeg only classifications performs generally poorly (average accuracy ~30% vs 25% pure chance); this may be due to inherent challenges of the signal itself:

- Signal to noise ratio – the electrodes may pick up undesired fluctuation
- Inter subject variability – anatomy and emotional response differences
- Session drift
- Feature scope – we used only 2 kinds of features
- Class overlap – emotions cannot be discretized reliably

Despite these issues, performance on second session remains consistent with initial model evaluation, indicating robust (despite being limited) generalization.

Eye Dataset

Features number	Eye movement parameters	Extracted features
1-12	Pupil diameter (X and Y)	Mean, standard deviation and DE features in four bands
13-16	Dispersion (X and Y)	Mean, standard deviation
17-18	Fixation duration (ms)	Mean, standard deviation
19-22	Saccade	Mean, standard deviation of saccade duration (ms) and saccade amplitude (°)
23-31	Event statistics	Blink frequency, fixation frequency, maximum fixation duration, total fixation dispersion, maximum fixation dispersion, saccade frequency, average saccade duration, average saccade amplitude, average saccade latency

df

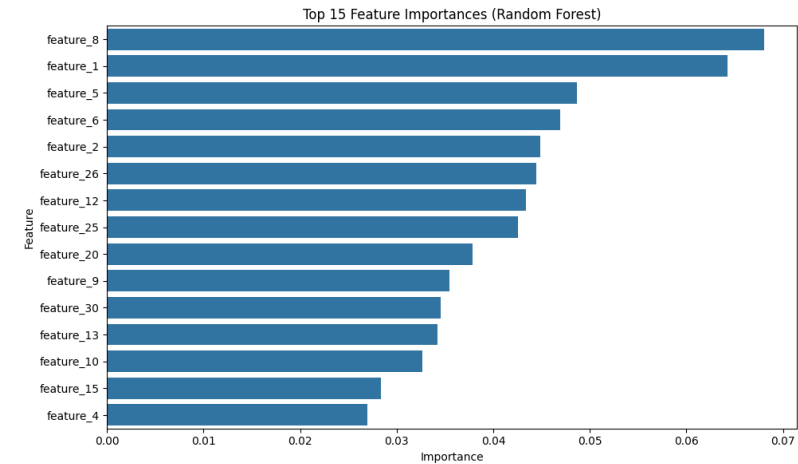
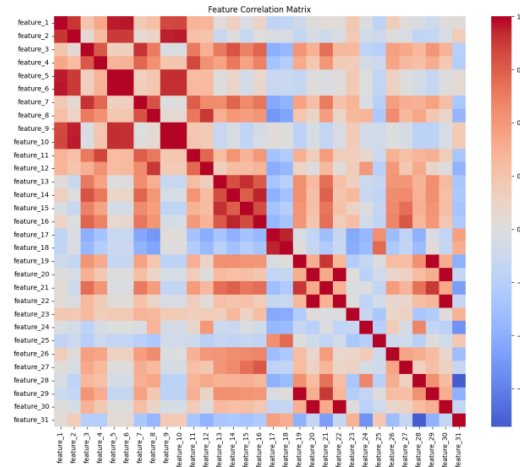
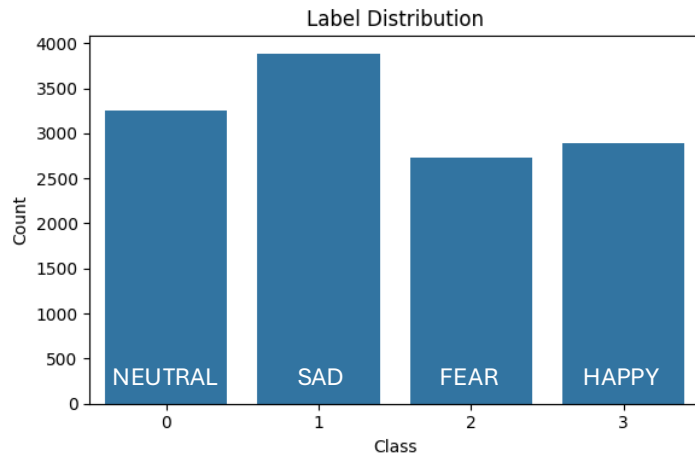
	feature_1	feature_2	feature_3	feature_4	feature_5	feature_6	feature_7	feature_8	feature_9	feature_10	...	feature_25	feature_26	feature_27	feature_28
0	73.827209	59.319793	3.408343	5.327512	24.379940	24.710870	13.399284	9.543863	23.766592	24.102732	...	2562.0	6149.1	115.9	1.0
1	73.826211	59.318841	3.407931	5.327245	24.379945	24.710877	13.398680	9.540437	23.766604	24.102751	...	2562.0	6149.1	115.9	1.0
2	73.820849	59.313086	3.408243	5.325591	24.379767	24.710709	13.399264	9.540427	23.766441	24.102621	...	2562.0	6149.1	115.9	1.0
3	73.809122	59.302406	3.408549	5.327105	24.379303	24.710260	13.399954	9.540258	23.766049	24.102268	...	2562.0	6149.1	115.9	1.0
4	73.799325	59.290777	3.407096	5.330190	24.378905	24.709867	13.398809	9.538985	23.765605	24.101868	...	2562.0	6149.1	115.9	1.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
12760	88.722452	58.335997	6.732136	6.519401	24.915197	25.249491	15.180500	13.887176	23.693112	24.026487	...	1031.6	10104.0	112.6	3.0
12761	88.750749	58.357305	6.737688	6.531835	24.916139	25.250446	15.182353	13.887315	23.694141	24.027557	...	1031.6	10104.0	112.6	3.0
12762	88.770440	58.370698	6.745927	6.543427	24.916855	25.251188	15.184108	13.887475	23.694821	24.028273	...	1031.6	10104.0	112.6	3.0
12763	88.781107	58.378891	6.752821	6.550072	24.917270	25.251612	15.183916	13.888405	23.695175	24.028598	...	1031.6	10104.0	112.6	3.0
12764	88.784180	58.382803	6.760918	6.554757	24.917451	25.251797	15.184306	13.890108	23.695403	24.028795	...	1031.6	10104.0	112.6	3.0

12765 rows x 34 columns

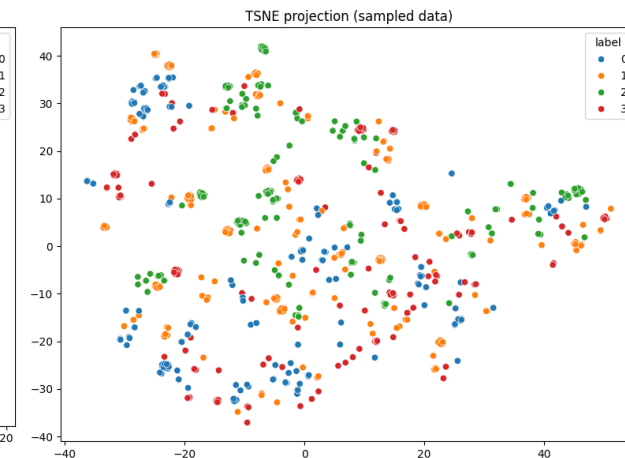
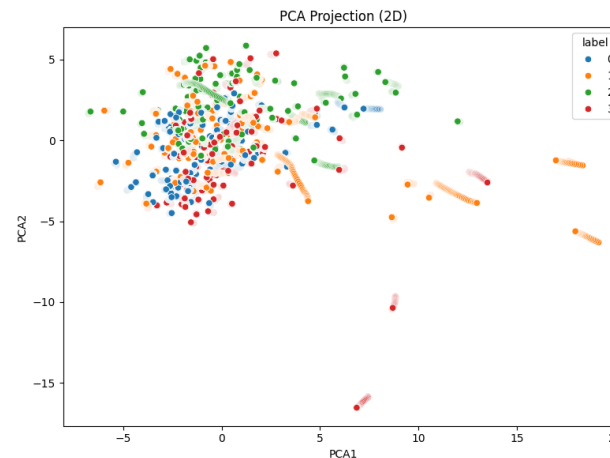
EYE DATASET ORGANIZATION

Data have been loaded and organized in a DataFrame of **12765 rows** (features computed for samples obtained taking 4 seconds windows from the recordings of the eye tracking glasses) and 34 columns: **31 features** – described in the table – and three columns for subject, number of trial and target emotion.

Eye Dataset Exploratory Analysis



- No missing or NaN values
- Classes are a bit unbalanced (*sad* is more represented)
- Subjects have the same number of samples
- Many observations show similar characteristics in PCA projection (but e.g. for class 3 there are distinguishable patterns)
- The t-SNE projection reveals some clusters and captures non-linear relationships in the data.
- The correlation matrix reveals the presence of strongly correlated features, indicating possible redundancy, but also uncorrelated features that might carry unique independent information
- The RF model identifies a subset of features effective in distinguishing between the target classes.



Eye Movement Classification

FIRST APPROACH

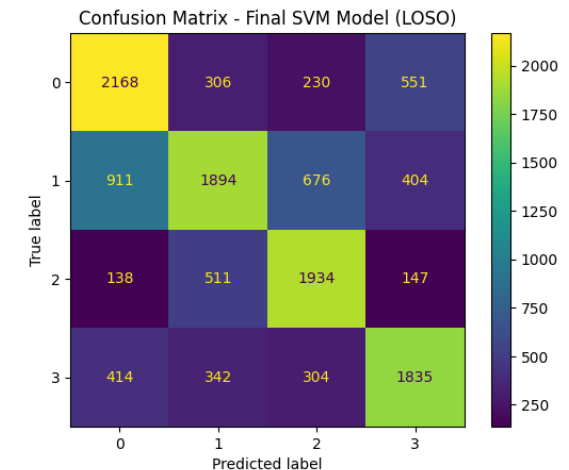
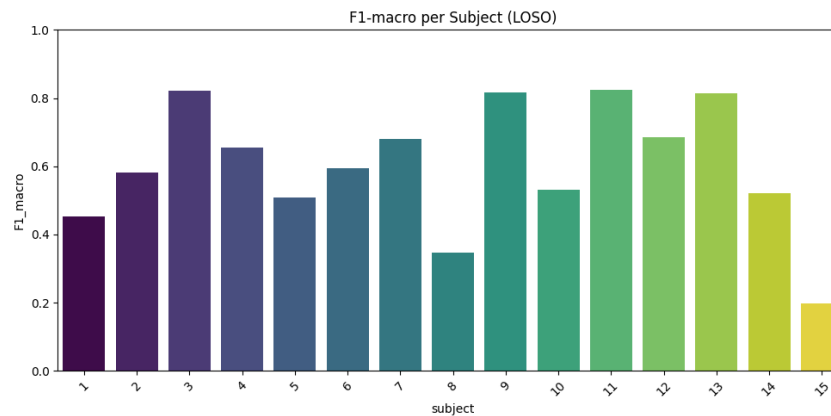
- Leave One Subject Out Validation
- Grid search parameters
- Comparison between Support Vector Machine, Random Forest Classifier, K-Nearest Neighbors
- SelectKBest for feature selection with score function `f_classif` (compute the ANOVA F-value for the provided sample)
 - The best model is SVM, with the following performances in LOSO validation.

Best parameters: {'select_k': 25, 'svc_C': 0.8, 'svc_gamma': 0.015}
Best F1-macro score (CV): 0.6017558113750587

=== Classification Report ===

	precision	recall	f1-score	support
0	0.597	0.666	0.630	3255
1	0.620	0.488	0.546	3885
2	0.615	0.708	0.658	2730
3	0.625	0.634	0.629	2895
accuracy			0.613	12765
macro avg	0.614	0.624	0.616	12765
weighted avg	0.614	0.613	0.610	12765

Final F1-macro (LOSO): 0.6158609602302387



Eye Movement Classification

FIRST APPROACH – Test on Second Session Dataset

To avoid data leakage, since for LOSO CV the models have already seen the whole dataset, they have been tested also on the second session dataset, that is completely new and independent.

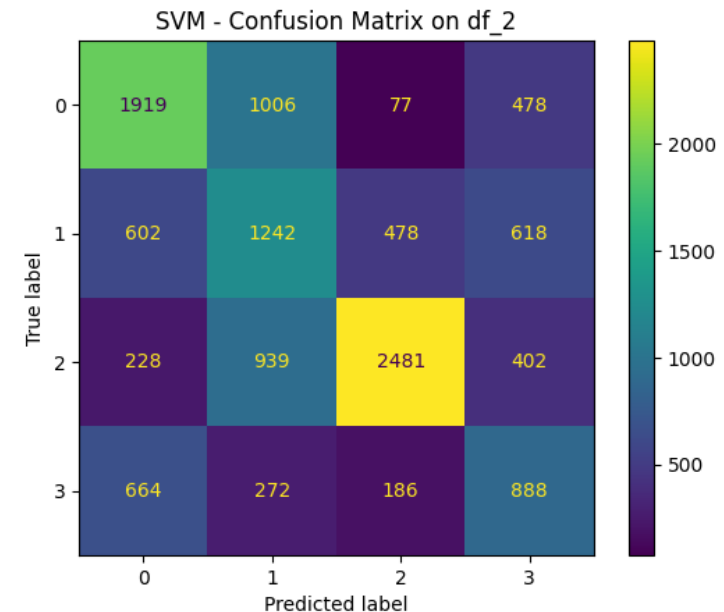
SVM confirms the best performances, even if they are obviously worse with respect to the previous case.

This is absolutely understandable and confirms that the model is able to generalize to new data.

```
=== SVM on df_2 ===
```

	precision	recall	f1-score	support
0	0.562	0.551	0.557	3480
1	0.359	0.422	0.388	2940
2	0.770	0.613	0.682	4050
3	0.372	0.442	0.404	2010
accuracy			0.523	12480
macro avg	0.516	0.507	0.508	12480
weighted avg	0.551	0.523	0.533	12480

F1-macro SVM (df_2): 0.5078323195826797



Eye Movement Classification

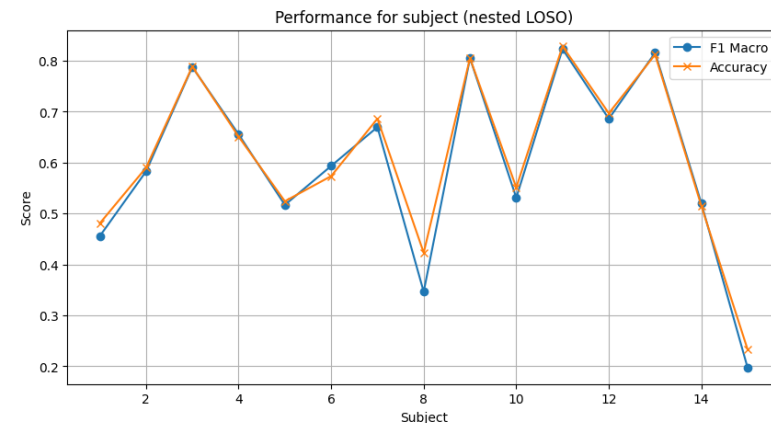
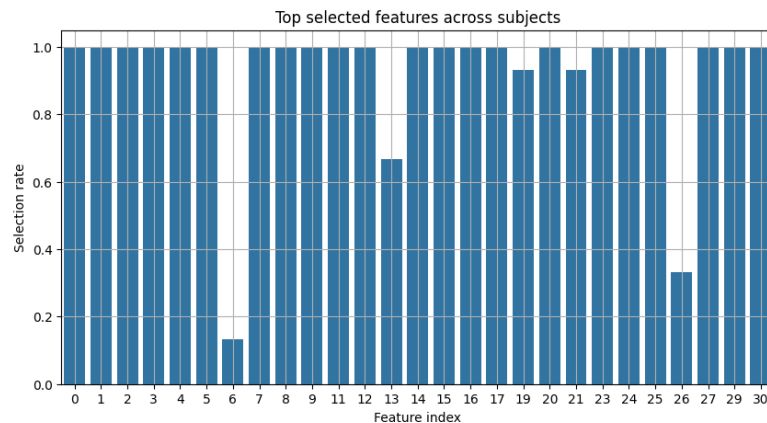
SECOND APPROACH: Nested GridSearch – Nested LOSO models by subject

Model selection without nested CV uses the same data to tune model parameters and evaluate model performance. Information may thus “leak” into the model and overfit the data. To avoid this problem, nested CV effectively uses a series of train/validation/test set splits. In the inner loop (executed by `GridSearchCV`), the score is approximately maximized by fitting a model to each training set, and then directly maximized in selecting (hyper)parameters over the validation set. In the outer loop, generalization error is estimated by averaging test set scores over several dataset splits [1].

All the models for each subject selected $k = 15$ as optimal number of features, selected as in the histogram below.

```
pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("select", SelectKBest(score_func=f_classif)),
    ("svc", SVC(kernel="rbf", class_weight="balanced"))
])
```

```
param_grid = {
    "select_k": [10, 15, 20, 25],
    "svc_C": [0.8, 0.9, 1.0, 1.2],
    "svc_gamma": [0.015, 0.017, 0.02]
}
```



[1] https://scikit-learn.org/stable/auto_examples/model_selection/plot_nested_cross_validation_iris.html

Eye Movement Classification

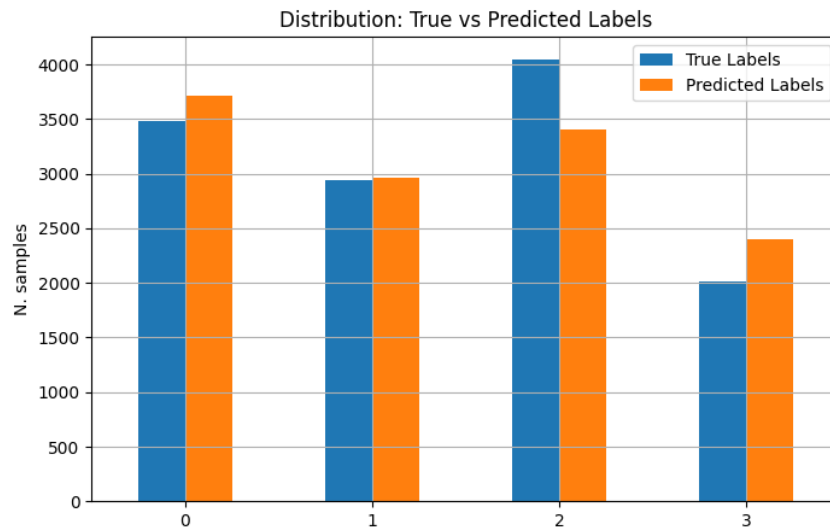
SECOND APPROACH: Nested GridSearch – Final model and test on second session

The final model has these parameters:

- $k \rightarrow$ mode of the k values of the models for each subject
- $C \rightarrow$ mean of the C values of the models for each subject
- $\gamma \rightarrow$ mean of the γ values of the models for each subject

It's fitted on the whole dataset and tested on the second independent session.

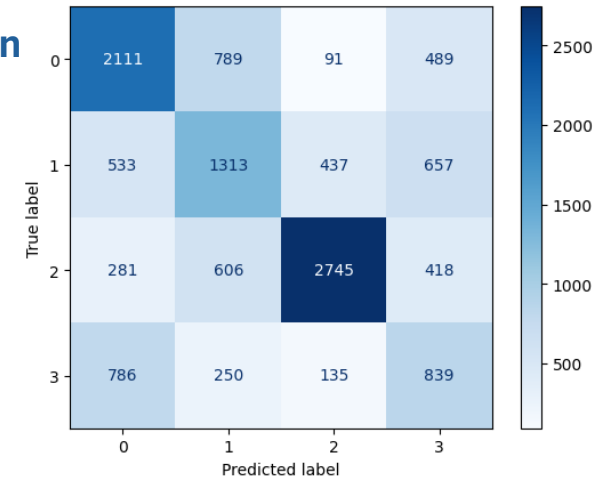
The Pipeline includes also `StandardScaler`.



Low precision and recall: confusion with other classes

Best performance

	precision	recall	f1-score	support
0	0.5688	0.6066	0.5871	3480
1	0.4439	0.4466	0.4452	2940
2	0.8055	0.6778	0.7361	4050
3	0.3491	0.4174	0.3802	2010
accuracy			0.5615	12480
macro avg	0.5418	0.5371	0.5372	12480
weighted avg	0.5808	0.5615	0.5687	12480



Eye Movement Classification

feature_name	f_score
feature_1	2091.836133
feature_6	1712.490879
feature_5	1660.953960
feature_2	1501.310281
feature_10	1183.946981
feature_9	1148.940116
feature_8	469.730602
feature_24	360.825404
feature_12	321.718309
feature_25	290.749826
feature_26	230.806353
feature_28	180.449781
feature_31	164.944609
feature_17	146.992107
feature_18	131.716017
feature_4	127.104796
feature_3	121.258006
feature_21	81.093394
feature_13	76.498716
feature_20	74.846255
feature_30	74.710453
feature_15	55.215419
feature_22	54.785410
feature_16	52.858757
feature_14	44.794498

THIRD APPROACH: Hybrid Feature Selection

The model selects these 25 features, sorted by `f_score`.

F-score feature selection: method used to rank and select relevant features based on their scores that represent the ratio of variability between groups (explained variance) to the variability within groups (unexplained variance). It evaluates features individually, so it's a univariate method.

To improve the model performance, also `mutual_info_classif` is considered to:

- capture non-linear relationships: MI can identify dependencies that correlation might miss.
- provide a clear ranking of feature importance based on their relevance to the target.
- Identify and reduce redundancy between features.

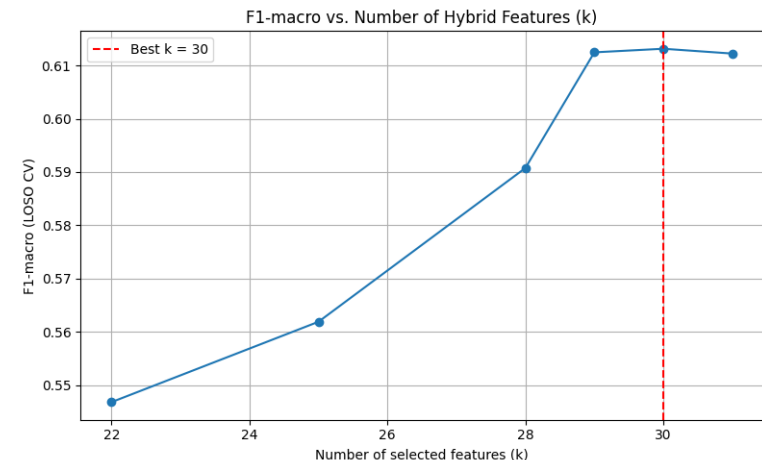


→ Hybrid feature selection strategy was applied by combining the normalized scores from both ANOVA F-test (`f_classif`) and mutual information (`mutual_info_classif`), capturing both linear and nonlinear relevance to the target classes.

→ Grid search to identify the optimal number of top-ranked hybrid features (`k`) using LOJO cross-validation, chosen in this list [22, 25, 28, 29, 30, 31].

→ For each value of `k`, a Support Vector Machine (SVM) with RBF kernel was trained using fixed hyperparameters previously obtained from nested cross-validation.

→ The best value of `k` is `k = 30`.



Eye Movement Classification

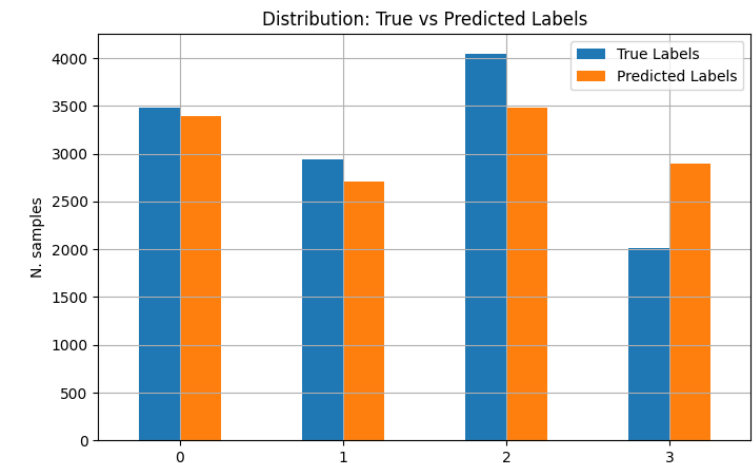
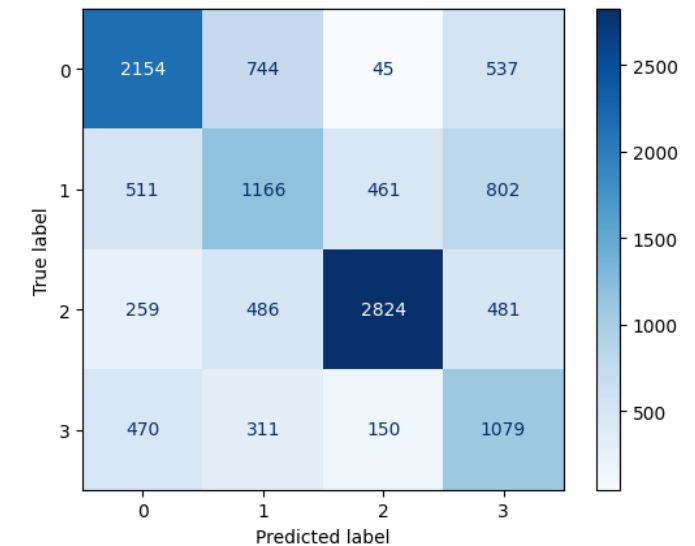
THIRD APPROACH: Hybrid Feature Selection – Test on second session

Compared to the baseline SVM model with `SelectKBest` ($F1 \approx 0.537$), the hybrid-selected model improved generalization by approximately 2% in macro F1, without sacrificing interpretability:

- more robust and generalizable model
- effective in multi-subject classification settings

- Accuracy: 57,88%
- F1: 55,73%
- Class 2: easiest to classify
- Class 3: largest performance gain
- Class 1: slight degradation → residual overlap in feature space

	precision	recall	f1-score	support
0	0.6346	0.6190	0.6267	3480
1	0.4307	0.3966	0.4130	2940
2	0.8115	0.6973	0.7501	4050
3	0.3722	0.5368	0.4396	2010
accuracy			0.5788	12480
macro avg	0.5623	0.5624	0.5573	12480
weighted avg	0.6017	0.5788	0.5863	12480

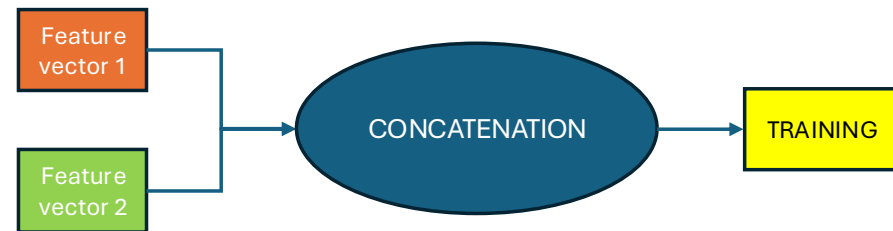


Multimodal Classification: Early Fusion

Early fusion, or **feature-level fusion**, is a data integration approach where multiple data sources are combined at the feature level before being fed into the machine learning model. Raw data from different modalities are processed and merged into a single feature set, which is then used for training the model [2].

HOW IT WORKS

- Feature extraction from each modality
- Concatenation: features are concatenated into a single feature vector
- Training: the concatenated vector is used to train the ML model



df_fusion_roi

	power_Prefrontal_delta	de_Prefrontal_delta	power_Prefrontal_theta	de_Prefrontal_theta	power_Prefrontal_alpha	de_Prefrontal_alpha	power_Prefrontal_beta	de_Prefrontal_beta
0	5.419316	2.788496	0.965721	1.348985	0.285273	0.536918	0.102753	0.556281
1	5.334861	2.772089	1.120173	1.496136	0.199796	0.172896	0.118957	0.700144
2	6.035457	2.896235	0.885443	1.264420	0.135681	-0.212997	0.048029	-0.202956
3	6.615760	2.988065	0.577991	0.838034	0.115213	-0.369540	0.026144	-0.813659
4	6.174299	2.918550	0.369509	0.387299	0.153445	-0.115044	0.094297	0.401473
...	...	...	...	...	...	...	...	...
12760	1.937747	1.758818	0.719483	1.055710	0.447947	0.988666	0.467953	2.072586
12761	5.407810	2.786447	1.336944	1.675631	0.183453	0.093695	0.040306	-0.384685
12762	4.383929	2.569521	0.783946	1.137337	0.606540	1.280543	0.143502	0.872944
12763	5.543900	2.810762	1.359158	1.691466	0.136645	-0.206578	0.049256	-0.183756
12764	4.787874	2.660478	1.019307	1.403549	0.338062	0.701286	0.151569	0.896582

12765 rows × 94 columns

ADVANTAGES

- Rich feature representation
- Simplicity
- Single training process

[2] <https://www.geeksforgeeks.org/deep-learning/early-fusion-vs-late-fusion-in-multimodal-data-processing/>

Multimodal Classification: Early Fusion

MODEL SELECTION

- Three models tested: SVM, Random Forest, Logistic Regression
- Leave One Subject Out Cross Validation
- Feature Selection: no selection, SelectKBest, L1-based, RandomForest Importance
- Standard Scaler

Nested LOSO Final Report:

	precision	recall	f1-score	support
0	0.66	0.76	0.70	3255
1	0.59	0.40	0.47	3885
2	0.61	0.76	0.68	2730
3	0.62	0.64	0.62	2895
accuracy			0.62	12765
macro avg	0.62	0.64	0.62	12765
weighted avg	0.62	0.62	0.61	12765

LOS0 performance

Nested LOSO Mean Accuracy: 62.07%

	precision	recall	f1-score	support
0	0.55	0.54	0.55	3480
1	0.38	0.40	0.39	2940
2	0.76	0.74	0.75	4050
3	0.39	0.40	0.39	2010
accuracy			0.55	12480
macro avg	0.52	0.52	0.52	12480
weighted avg	0.55	0.55	0.55	12480

Performance on 2° session

BEST MODEL

SVM (RBF) + SelectKBest (F-test) - LOSO Mean Accuracy: 62.77%				
	precision	recall	f1-score	support
0	0.64	0.76	0.70	3255
1	0.59	0.41	0.48	3885
2	0.62	0.75	0.68	2730
3	0.65	0.65	0.65	2895
accuracy			0.63	12765
macro avg	0.63	0.64	0.63	12765
weighted avg	0.62	0.63	0.62	12765

HYPERPARAMETER TUNING

Nested GridSearchCV:

- $k = 67$
- $C = 8$
- $\text{gamma} = 0.00075$

The performance is a bit worse because Nested GridSearchCV provides a more reliable estimation avoiding data leakage.

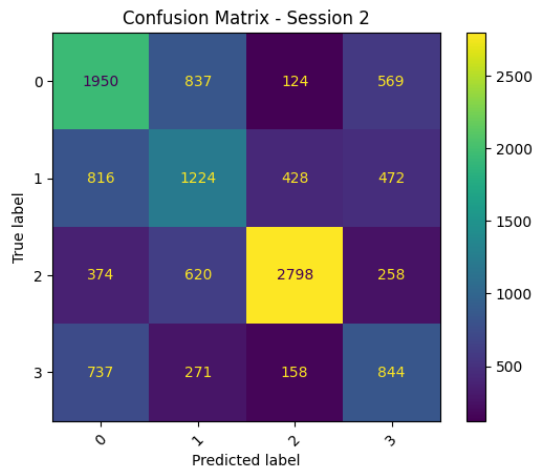
Multimodal Classification: Early Fusion

EVALUATION OF THE PERFORMANCE VARYING k

- C and gamma are kept fixed
- SelectKBest

BEST MODEL: k = 29

Performance on 2° session

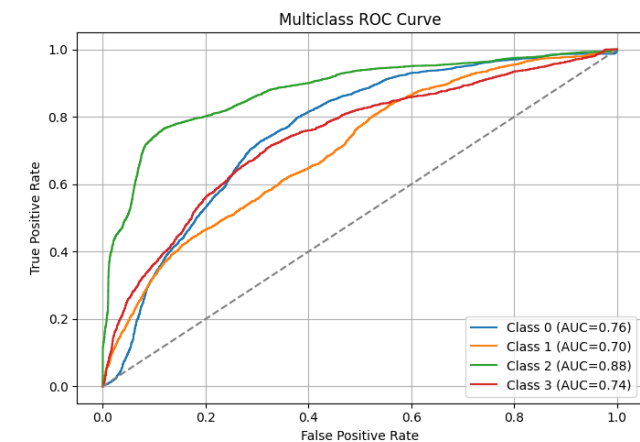
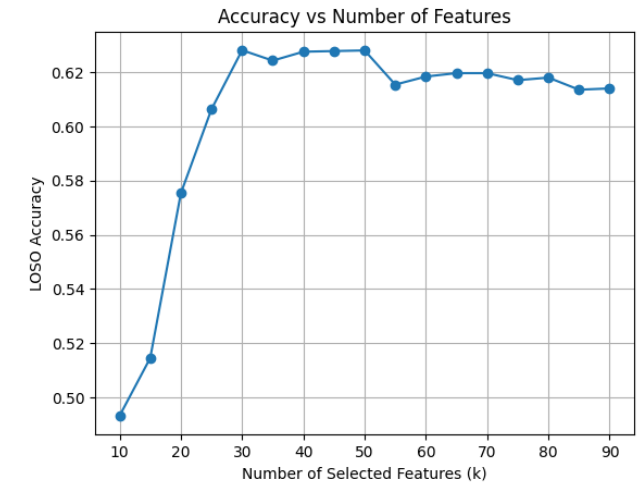


	precision	recall	f1-score	support
0	0.50	0.56	0.53	3480
1	0.41	0.42	0.42	2940
2	0.80	0.69	0.74	4050
3	0.39	0.42	0.41	2010
accuracy			0.55	12480
macro avg	0.53	0.52	0.52	12480
weighted avg	0.56	0.55	0.55	12480

Selected features:

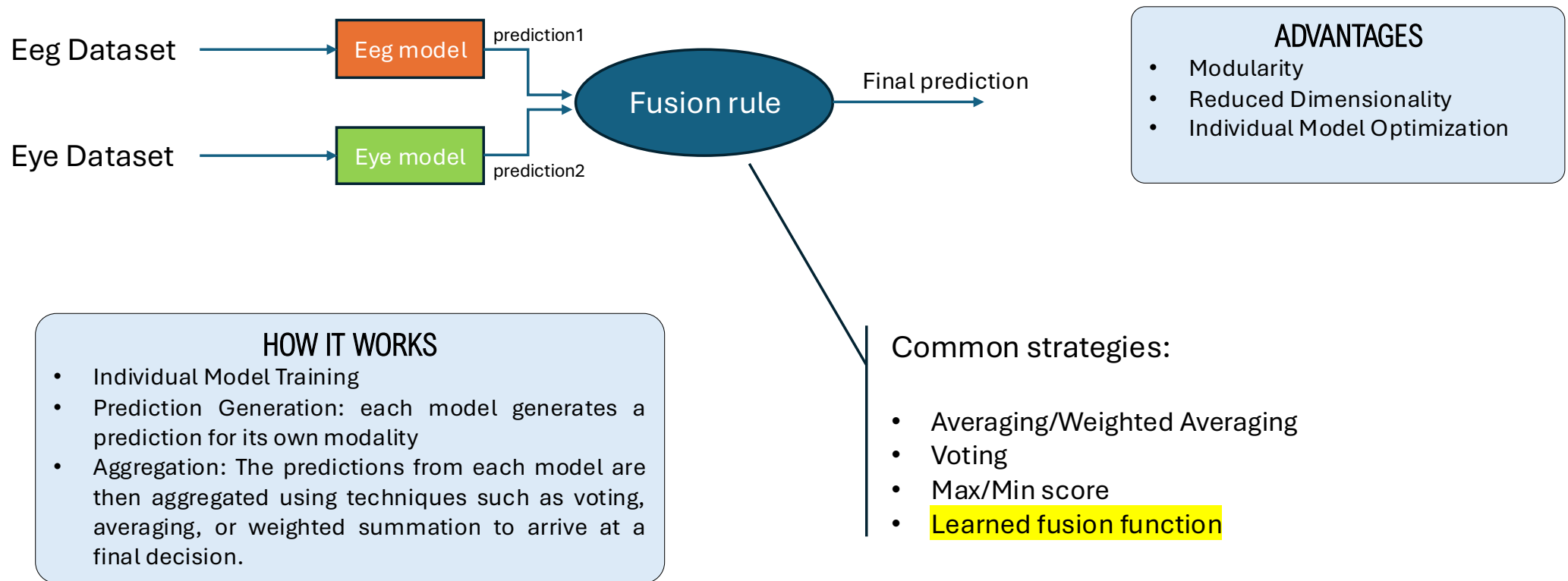
```
Index(['power_Temporal_delta', 'de_Temporal_delta', 'de_Temporal_theta',
      'power_Temporal_beta', 'de_Temporal_beta', 'power_Temporal_gamma',
      'de_Temporal_gamma', 'feature_1', 'feature_2', 'feature_3', 'feature_4',
      'feature_5', 'feature_6', 'feature_8', 'feature_9', 'feature_10',
      'feature_12', 'feature_13', 'feature_15', 'feature_17', 'feature_18',
      'feature_20', 'feature_21', 'feature_24', 'feature_25', 'feature_26',
      'feature_28', 'feature_30', 'feature_31'],
      dtype='object')
```

- This model performs slightly better on classes 1 and 3
- The previous one is marginally better on class 0
- Class 2 shows the highest discriminability, as evidenced by the largest AUC
- Class 1 has the lowest AUC
- The performance on Classes 0 and 3 is moderate but still better than random guessing (AUC > 0.5)



Multimodal Classification: Late Fusion

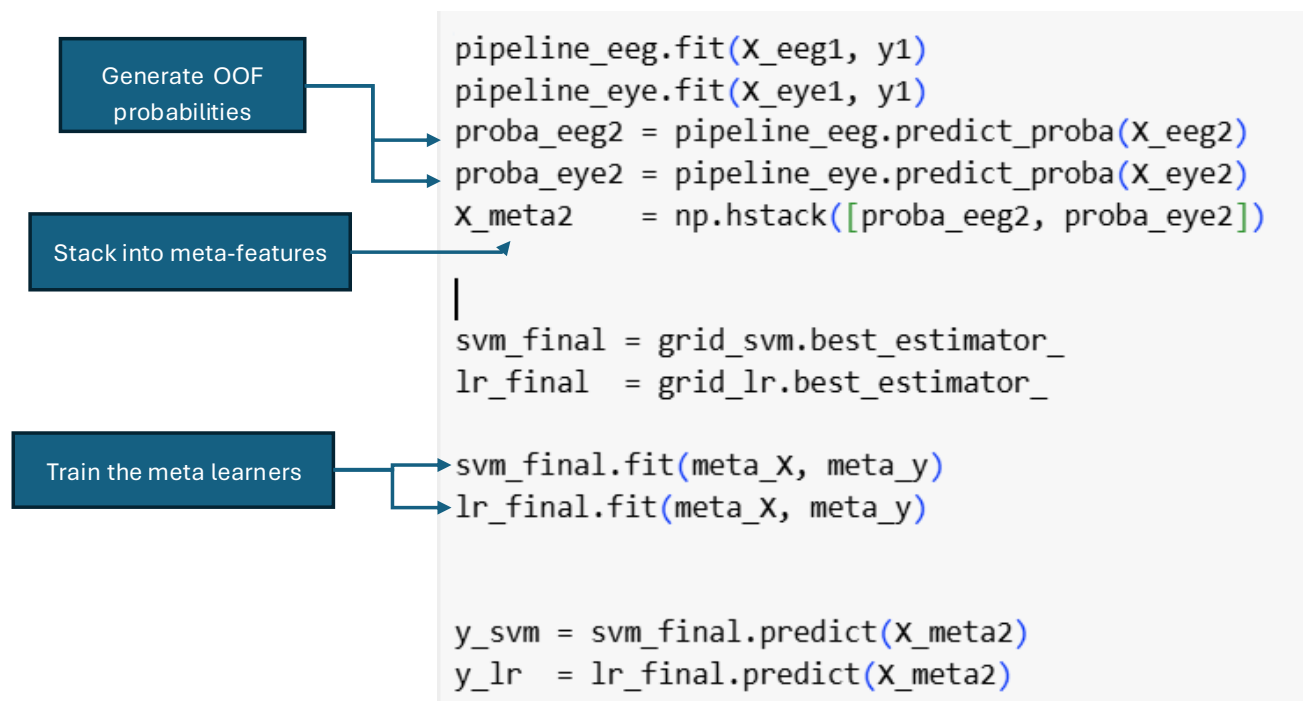
Late fusion is an approach in which data from each modality is processed independently through separate models, and the outputs from these models are then combined at a later stage.



Multimodal Classification: Late Fusion

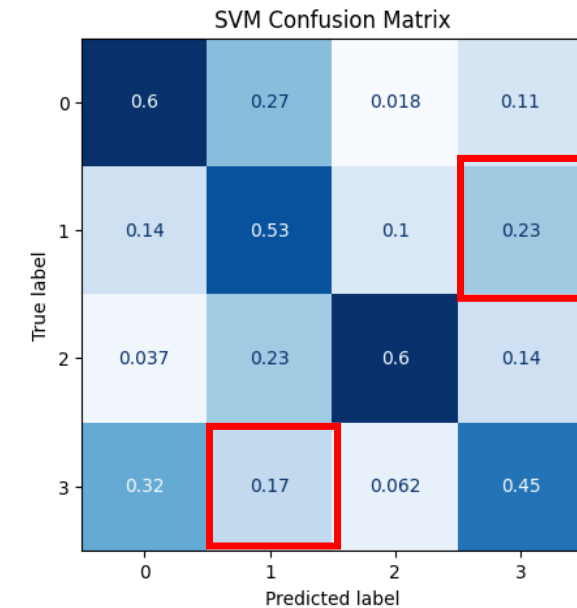
Stacking is a machine learning strategy that combines the predictions of numerous base models, also known as first-level models or base learners, and feeds them into a higher-level model, also known as a meta-model to make the final prediction.

- Meta-learner types: SVM & Logistic Regression
- Hyperparameter tuning: Grid search over C, γ and penalty/weights for LR
- Evaluation: LOSO cross-validation on Session 1 $\rightarrow$ select best model
- Retrain on all Session 1 (training done on stacked probabilities)
- Test on held-out Session 2



Multimodal Classification: Late Fusion

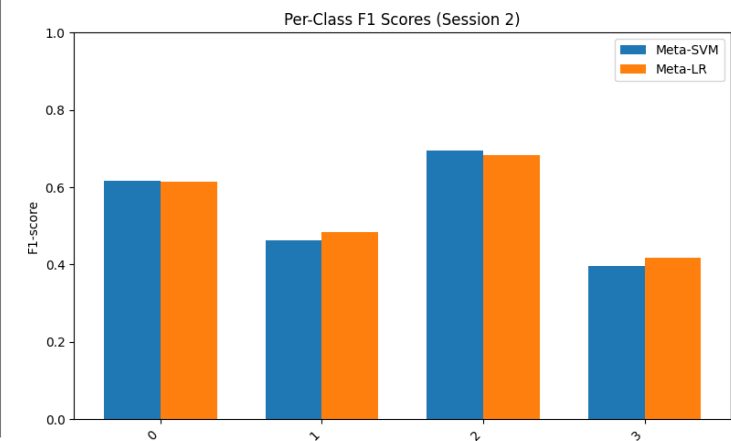
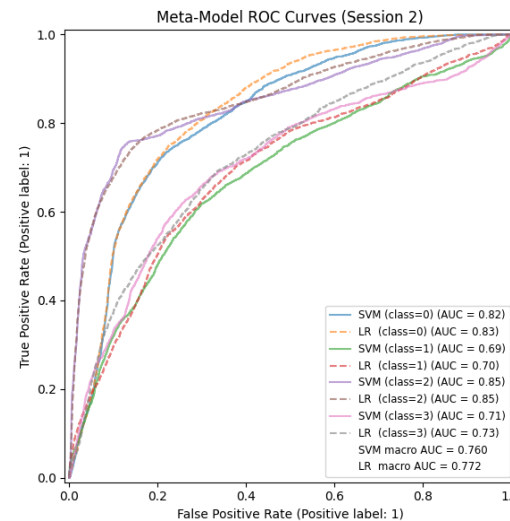
- Both meta-models deliver similar macro-F1 (SVM 54.2 % vs. LR 55.0 %) and accuracy (55.7 % vs. 56.1 %).
- Class 0 (Neutral) achieves the highest recall (~60 %) for both models.
- Class 3 (Happy) remains the hardest, often misclassified as Class 1 (Sad).
- Meta-LR marginally outperforms on macro-F1 and per-class F1 for Sad and Happy, but the gains are very small.



=== SVM Meta-Model Test (Session 2) ===

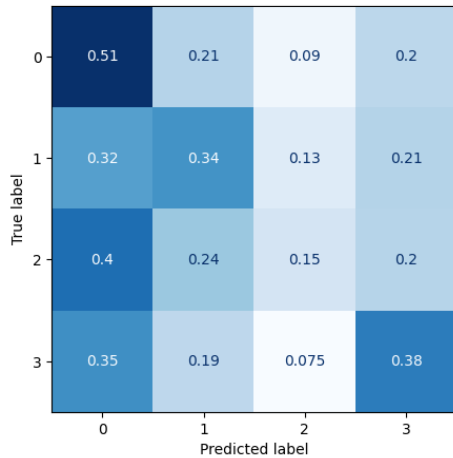
	precision	recall	f1-score	support
0	0.634	0.599	0.616	3480
1	0.412	0.525	0.462	2940
2	0.832	0.597	0.695	4050
3	0.355	0.448	0.396	2010
accuracy			0.557	12480
macro avg	0.558	0.542	0.542	12480
weighted avg	0.601	0.557	0.570	12480

Accuracy: 0.556650641025641

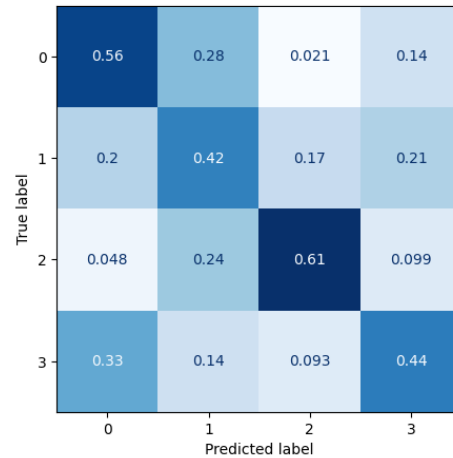


Models Comparison

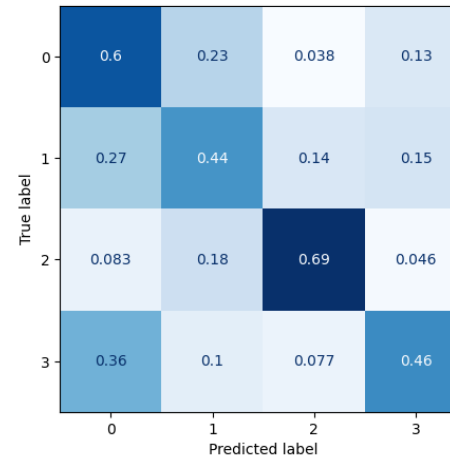
EEG-only Confusion Matrix (Normalized)



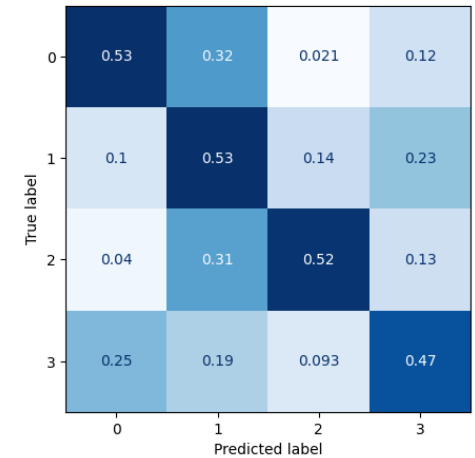
Eye-only Confusion Matrix (Normalized)



Early-fusion Confusion Matrix (Normalized)



Late-fusion Confusion Matrix (Normalized)



Wilcoxon signed-rank test results (LOSO F1-macro):

Late vs Early : $W=47.0$, $p=0.7299 \rightarrow \text{n.s.}$

Late vs Eye-only : $W=49.0$, $p=0.5318 \rightarrow \text{n.s.}$

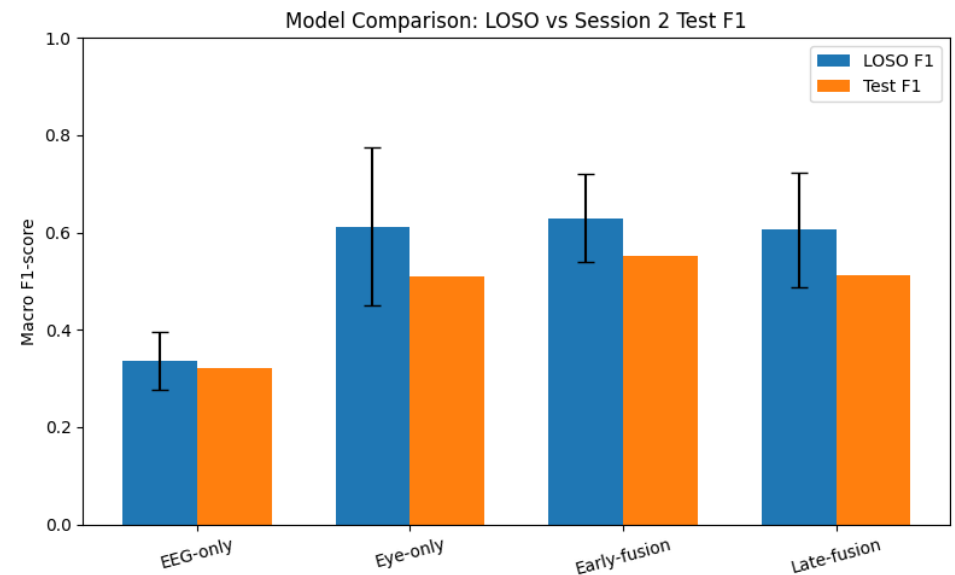
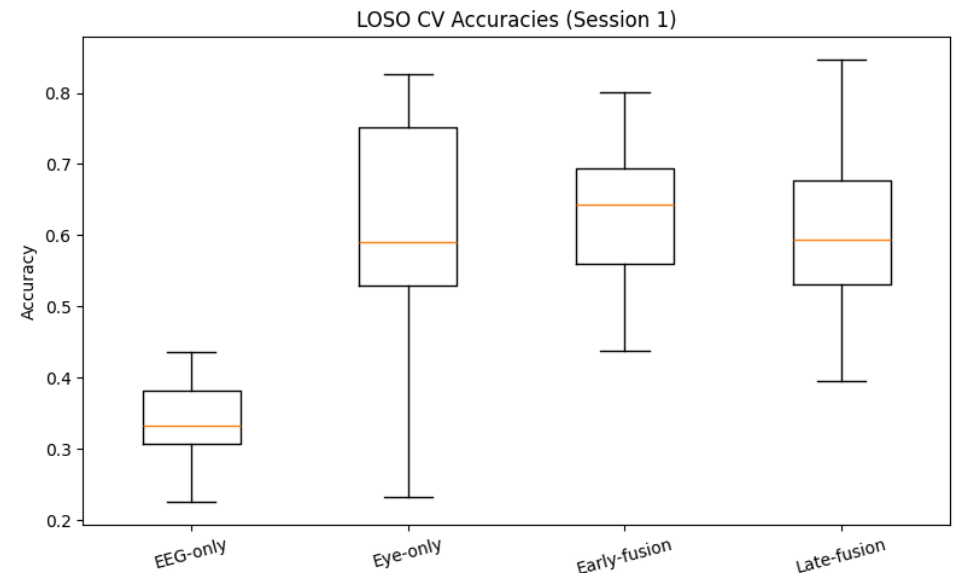
Early vs Eye-only : $W=54.0$, $p=0.7615 \rightarrow \text{n.s.}$

Late vs Eeg-only : $W=1.0$, $p=0.0001 \rightarrow \text{significant}$

Early vs Eeg-only : $W=0.0$, $p=0.0001 \rightarrow \text{significant}$

Models Comparison

- **EEG-only underperforms:**
Lowest median accuracy and narrowest inter-quartile range → least reliable
- **Eye-only is inconsistent:**
Mean accuracy on par with fusion methods but with much higher variance.
- **Fusion methods stabilize performance:**
Both early-fusion and late-fusion achieve similar median accuracies and lower variance than eye-only.
- **No difference among fusions and Eye-only:**
There's no significant gap between Late vs. Early fusion ($p = 0.73$) or between either fusion method and the eye-only model ($p > 0.5$), indicating that eye-tracking is the dominant modality and EEG adds virtually no boost when fused



Comparison with literature

- While our LOSO CV on Session 1 is truly subject-independent, our final hold-out test on Session 2 uses the same subjects (but different trials), so it assesses cross-session generalization rather than new-subject performance.
- EmotionMeter's 16/8 split is within the same session (subjects seen in both train and test), so it measures intra-session, subject-dependent accuracy. Our Session 2 test is strictly inter-session, which introduces additional sources variability.
- EEG-only, subject-independent accuracy: 37.99 % (Zhong et al.) vs. our 33.6 %. This may be due to a more thorough preprocessing on their part and a much wider search space for the parameter C

Subject-independent classification mean accuracy (Zhong et al.)

	SEED-IV
Model	all bands
SVM	37.99/12.52

Subject-dependent classification mean accuracy (Zhong et al.)

	SEED-IV
Model	all bands
SVM	56.61/20.05
EmotionMeter [24]	70.58/17.01

Model	SD Mean	SD Std	LOSO Mean	LOSO Std
EEG-only	0.381	0.063	0.336	0.060
Eye-only	0.459	0.175	0.612	0.162
Early-fusion	0.499	0.226	0.630	0.091
Late-fusion	0.654	0.209	0.605	0.117

Conclusions

Three classification pipelines developed and evaluated for multimodal emotion recognition using EEG signals and eye movement data: EEG-only, eye-only, multimodal fusion (early and late).

- Robust evaluation strategy
- Interpretability and transparency
- Extensive comparisons across modalities and across evaluation levels



Future improvements



Better EEG
preprocessing

Advanced fusion
techniques

Use of Neural
Networks

Broader
hyperparameters
search space

Richer feature
engineering

References

- Wei-Long Zheng, Wei Liu, Yifei Lu, Bao-Liang Lu, and Andrzej Cichocki, EmotionMeter: A Multimodal Framework for Recognizing Human Emotions. IEEE Transactions on Cybernetics, Volume: 49, Issue: 3, March 2019, Pages: 1110-1122, DOI: 10.1109/TCYB.2018.2797176.
- D. Nie, X.-W. Wang, L.-C. Shi, and B.-L. Lu, "EEG-based emotion recognition during watching movies," in Proc. 5th Int. IEEE/EMBS Conf. Neural Eng., 2011, pp. 667–670.
- P. Zhong, D. Wang and C. Miao, "EEG-Based Emotion Recognition Using Regularized Graph Neural Networks," in IEEE Transactions on Affective Computing, vol. 13, no. 3, pp. 1290-1301, 1 July-Sept. 2022, doi: 10.1109/TAFFC.2020.2994159.
- <https://bcmi.sjtu.edu.cn/~seed/seed-iv.html>